Amit Singh

Senior Backend Engineer | Python, AWS, Applied AI & Production Reliability

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Software engineer with 7+ years building and operating backend and platform systems across product safety, network operations and financial data. Primary depth in Python, AWS, event-driven workflows, relational data and production reliability; production experience with Claude, RAG, OpenSearch and human-in-the-loop review, plus hands-on Go projects. Known for resolving cross-service failures at their source, improving operator workflows, and owning delivery through rollout and support.

PROFESSIONAL EXPERIENCE

Amazon Ring (client) | Realtek Software Services (employer)

Software Development Engineer II · Ring Safety engineering · 2025–present · Pune, India

- Own backend, data and platform delivery for a multi-company product-safety system: shipped 10 production features, onboarded Blink and LEO, backfilled ~15K historical tickets, and support ~75K Ring and Blink records.
- Built Ring's PPM safety-metrics dashboard and a dedicated PPM Data Quality view, turning escalation and device-volume data into auditable reporting while surfacing missing mappings and denominator gaps before stakeholders see a number.
- Moved Ring's safety-ticket analytics dashboard from gamma to production in one day when a surge of incoming SIM tickets needed immediate reporting: exported the QuickSight data source, datasets, analysis and dashboard as an asset bundle, imported it with data-source overrides, fixed import-rejected filter identifiers and a dataset the update API refused by re-importing it alone, and loaded and row-count-verified the ~239K-row backing table before hand-off.
- Designed and delivered company-aware Project and Project Family capabilities across database writes, backend logic, tables, create/edit flows and ticket filtering; closed gating defects that let other-company values propagate into Lab126 and LEO workflows.
- Recovered failed Genie device-onboarding infrastructure, completed onboarding for five devices visible in the operational dashboard, and hardened the CloudFormation stack definitions and deployment checks afterwards.
- Migrated Claude inference after model deprecation and built text/image analysis with confidence-based human review; chose a conservative threshold that trades higher review volume for lower silent-misclassification risk, reducing estimated manual effort by ~30% for Ring and ~80% for Lab126.
- Root-caused a cross-service authorization regression to an identity-resolution change in a separately owned permission service; showed that multiple IAM/SigV4 paths shared the failure and drove the correction at the source rather than preserving a broken contract with a local workaround.
- Designed compliance-sensitive legal-hold recalculation across UI edits, asynchronous Jira events and scheduled reconciliation; isolated company-specific criticality behaviour and chose scheduled repair so state self-corrects after missed events.
- Restored a production-critical missing-data workflow within ~10 hours, then strengthened Aurora/CloudWatch monitoring and database-change controls with reviewed SQL, rollback plans, staged rehearsal and post-change verification. Mentor two engineers and serve in the on-call rotation.

Orange Business Services

Senior Software Engineer · Project-based engagement · 2024–2025

- Built and productionized Jenkins/Jira/Groovy pipelines for multi-vendor device integration and configuration updates, supporting integration, beta, user-acceptance and production rollout.
- Designed and productionized a ServiceNow-grounded RAG assistant using OpenSearch, embeddings and an LLM; reduced repetitive operational queries by an estimated ~50% with approximately ~80% response satisfaction.
- Led beta and user testing, added human verification for uncertain responses, and built a weekly knowledge-refresh lifecycle with review snapshots and controlled retention.

- Drove production readiness with integration testing, runbooks, customer guides and operational handover, using FastAPI/Django, AWS Lambda, Kubernetes, Redis and SQL optimization.

Virtusa Polaris | Lloyds Banking Group engagement

Software Engineer · 2019–2024

- Built APIs and ETL pipelines integrating Vermeg Colline collateral/risk data and Bloomberg market feeds for bonds, equities and mutual funds within defined regional SLA windows.
- Led PostgreSQL migration and Snowflake-integrated data pipelines; improved API response time by ~30% through Python optimization, SQL tuning and CI/CD improvements.
- Monitored production data pipelines with Dynatrace and managed incidents through ServiceNow, maintaining operational continuity across data dependencies.

SELECTED INDEPENDENT PROJECTS

FastAPI Microservices Platform | Python, PostgreSQL

Self-hosted webhook delivery on PostgreSQL alone, without Redis or Kafka: lease-fenced durable queue, Standard Webhooks signatures cross-verified against the reference library, SSRF-isolated egress and audited dead-letter replay; producer/receiver SDK and CLI on PyPI. A reproducible million-job correctness run recorded no lost jobs. Delivery remains at-least-once and the backlog-latency target was not met.

CloudScale Backend | Python, PostgreSQL

CQRS event-sourcing service with a typed hexagonal core: transaction-bound idempotency, balanced postings and a transactional outbox over SQLite and PostgreSQL adapters; default-deny authorization, rate limiting, append-only audit log, fail-closed migrations, leader-elected consumer failover, a STRIDE threat model and architecture decision records. An executable backup/restore drill rebuilds derived state. Recovery evidence is at seeded scale; sustained availability remains unproven.

Code Quality Analyzer | Python, published as cqa-analyzer

Eight-language static analyzer that runs offline by construction: a 62-pattern architecture catalog with honest not-applicable scoring, cross-file duplication and complexity, changed-line and config-pinned CI gates with SARIF output, and an MCP server so coding agents get the same results as CI. Scoring fairness is recalibrated against a public 24-project corpus before every policy change. Bounded static analysis, not proof of arbitrary program correctness.

AgentOS | Python, TypeScript

Durable agent workflows over an append-only event log with human approval gates, lease-based worker coordination and an operator UI with time-travel debugging; deterministic fault-injection tests at commit boundaries exercise crash recovery. External side effects depend on downstream idempotency or reconciliation.

Design decisions, benchmark scope and pinned CI evidence for every project: amitsinghom.github.io/evidence.html.

TECHNICAL SKILLS

Backend: Python, FastAPI, Django, SQLAlchemy, REST APIs, event-driven services, idempotency, transactional data processing; Go through hands-on projects.

AI systems: Amazon Bedrock, Claude, RAG, OpenSearch, embeddings and vector search, prompt/context optimization, confidence-gated human-in-the-loop workflows.

Cloud & operations: AWS Lambda, API Gateway, Step Functions, SQS/SNS, IAM, CloudFormation, CloudWatch, QuickSight, Docker, Kubernetes, Jenkins/Groovy, GitHub Actions, Dynatrace, incident response.

Data: SQL, PostgreSQL, Aurora MySQL, DynamoDB, Redis, Snowflake, schema design, migrations, query optimization, ETL.

Frontend & quality: React, TypeScript, Cloudscape; unit, integration and API testing, code review, runbooks.

EDUCATION

Bachelor of Engineering, Information Technology | University of Mumbai